

**Math 2551 Worksheet: Double Integrals on General Regions**

1. Compute the integrated integral

$$\int_0^\pi \int_0^x x \sin(y) \, dy \, dx.$$

2. Decide, without calculation, if each of the integrals below are positive, negative, or zero. Let  $D$  be the region inside the unit circle centered at the origin. Let  $T$ ,  $B$ ,  $R$ , and  $L$  denote the regions enclosed by the top half, the bottom half, the right half, and the left half of unit circle, respectively.

(a)  $\iint_B (y^3 + y^5) \, dA$

(d)  $\iint_L (y^3 + y^5) \, dA$

(b)  $\iint_T (y^3 + y^5) \, dA$

(c)  $\iint_D (y^3 + y^5) \, dA$

(e)  $\iint_R (y^3 + y^5) \, dA$

3. Write an iterated integral for  $\iint_R 1 \, dA$  over the region  $R$  using vertical cross-sections and horizontal cross-sections.

(a) Bounded by  $y = e^{-x}$ ,  $y = 1$ , and  $x = \ln 3$ .

(b) Bounded by  $y = x^2$  and  $y = x + 2$

4. Sketch the region of integration, reverse the order of integration, and evaluate the integral.

$$\int_0^{\sqrt{\pi}} \int_y^{\sqrt{\pi}} \cos(x^2) \, dx \, dy.$$

5. Sketch the region of integration and evaluate the integral

$$\iint_R xy^2 \, dA,$$

where  $R$  is enclosed by  $x = 0$  and  $x = \sqrt{1 - y^2}$ .

6. Find the volume of the solid bounded by the cylinder  $y^2 + z^2 = 4$  and the planes  $x = 2y$ ,  $x = 0$ ,  $z = 0$  in the first octant.
7. The integral expression below gives the area of a region in the  $xy$ -plane. Sketch the region, labeling the bounding curves with their equations, and giving the coordinates of points where the curves intersect. Then find the area of the region.

$$\int_0^2 \int_{x^2-4}^0 dy \, dx + \int_0^4 \int_0^{\sqrt{x}} dy \, dx$$